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SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal		
Instructor(s) Name	Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju		
CourseCode	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week11 – Wednesday**	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All batches
Assignment Number:21.3(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*Expected Time to complete*
1	**Lab 21 – Collaborative AI Coding: Pair Programming and Version Control Integration** **Lab Objectives:**	Week11 - Wednesday

- Learn to use AI as a pair programmer for writing, reviewing, and optimizing code.
- Understand how to integrate AI-generated code into version control systems (Git).

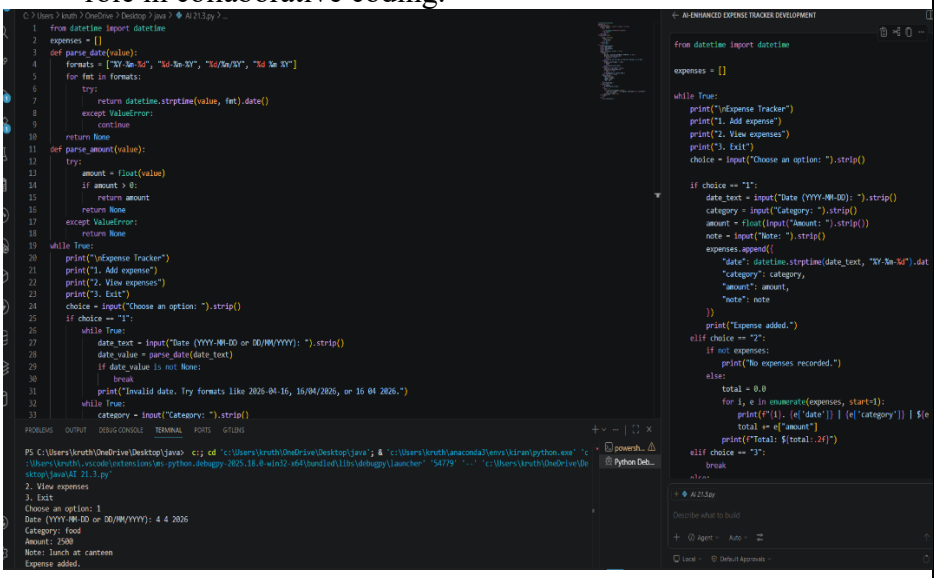
Practice collaborative workflows such as commit messages, branching, and merging.

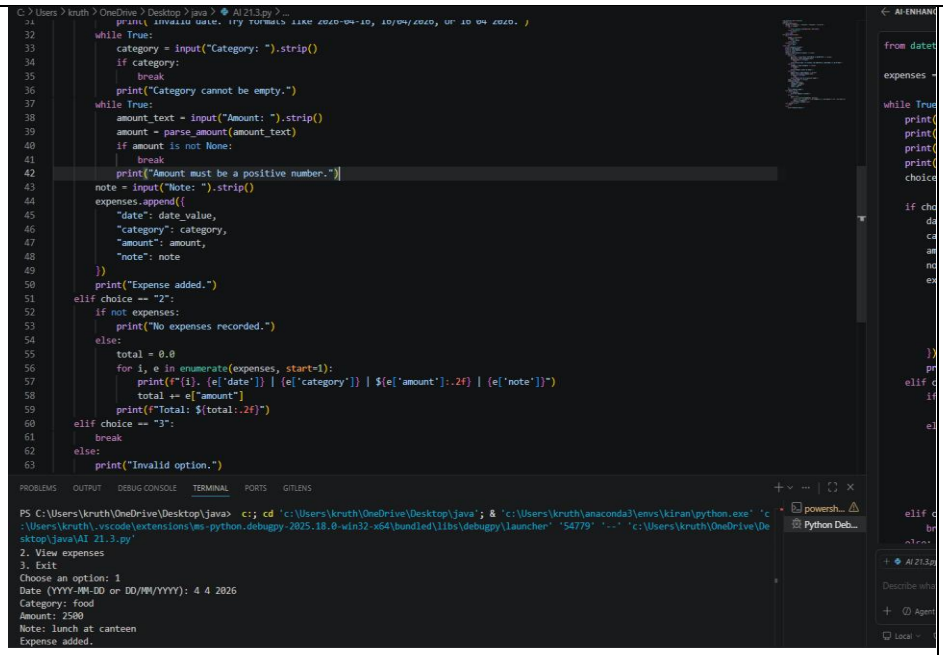
Task 1 – AI-Assisted Pair Programming

Example:

Use AI to generate an initial Expense Tracker Application.

- Ask AI to create a basic program to record daily expenses and categories.
- Prompt AI to suggest improvements such as:
  - Monthly expense summary
  - Input validation
  - Code refactoring into functions
- Compare the initial and improved versions to understand AI's role in collaborative coding.



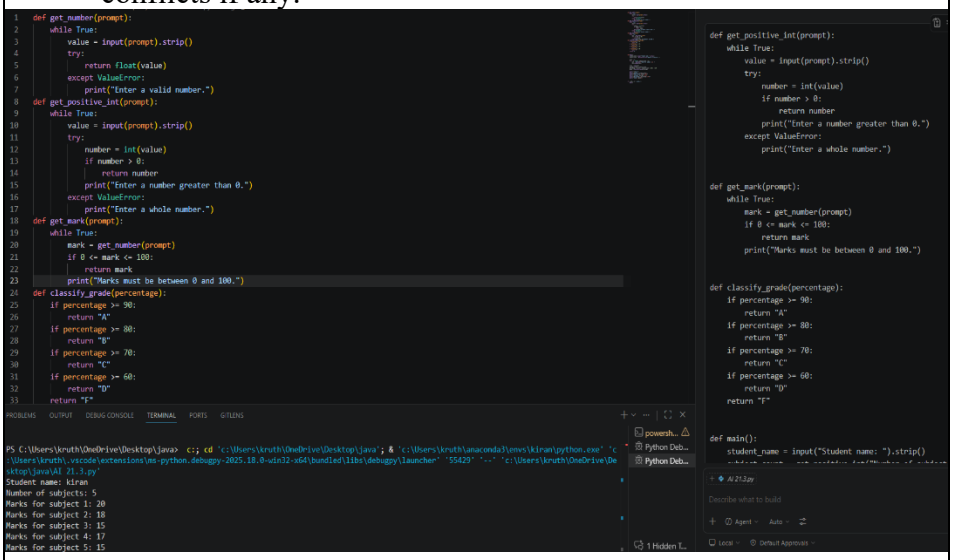


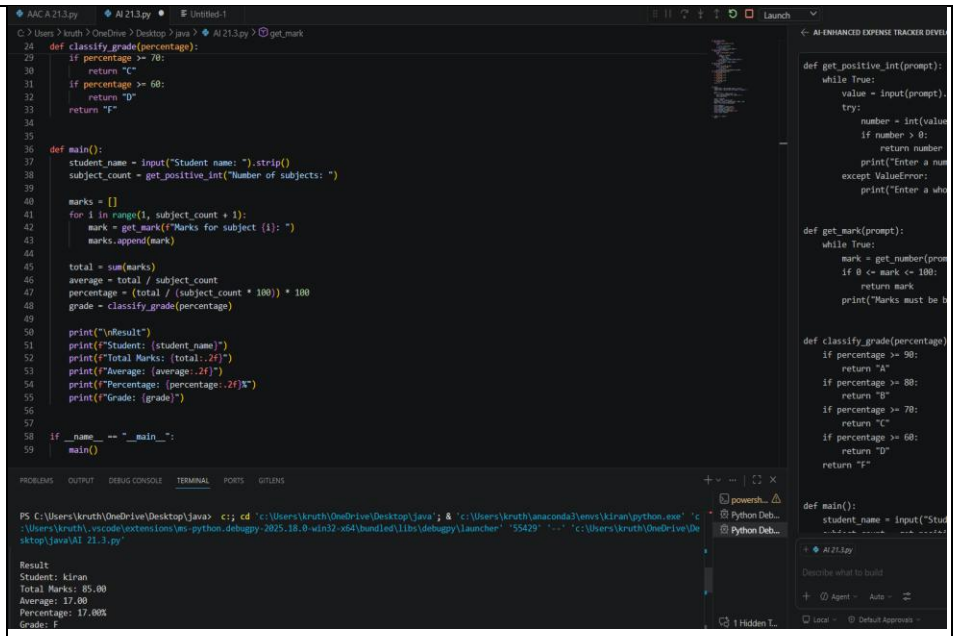
Task 2 – Version Control Integration

Example:

Manage a Student Grade Calculator project using Git.

- Initialize a Git repository for the project.
- Create a new branch to add AI-assisted features such as:
  - Grade classification (A, B, C, etc.)
  - Average and percentage calculation
- Commit changes with meaningful commit messages.
- Merge the feature branch back into the main branch and resolve conflicts if any.



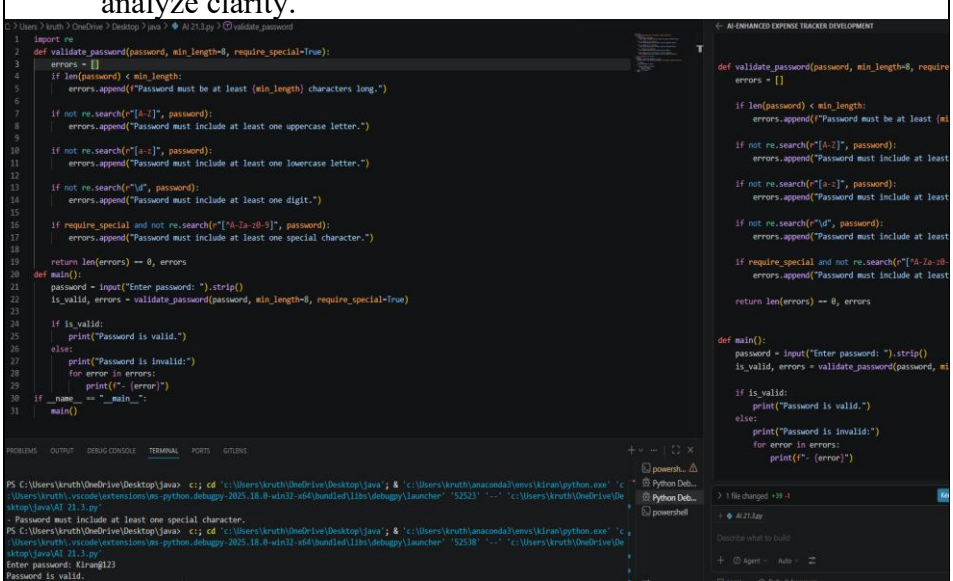


Task 3 – AI for Commit Message Generation

Example:

Use AI to generate commit messages for a Password Validation Module.

- Modify the code to add rules like:
  - Minimum length
  - Special character requirement
- Ask AI to generate commit messages describing these updates.
- Compare AI-generated messages with human-written ones and analyze clarity.

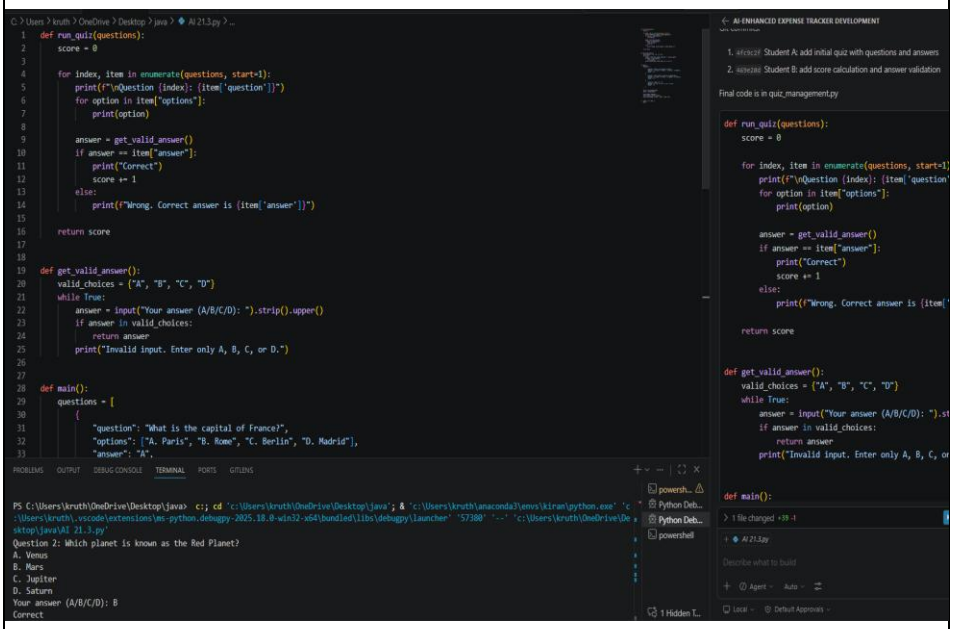


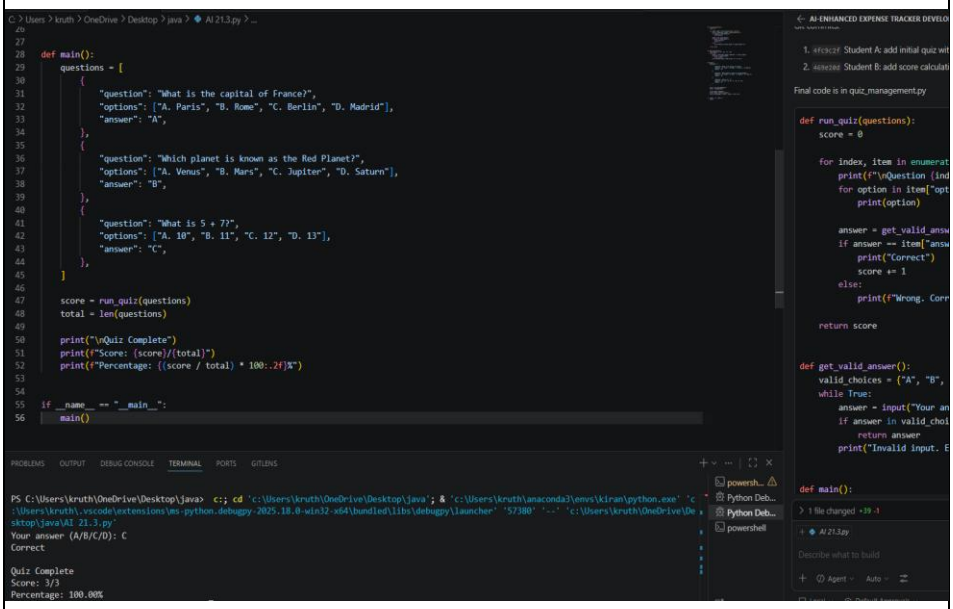
Task 4 – Pair Programming Simulation

Scenario:

Collaboratively develop a Quiz Management System.

- Student A creates the initial quiz program with questions and answers.
- Student B, with AI assistance, adds features such as:
  - Score calculation
  - Input validation
- Both students collaborate using Git branching and merging to integrate their changes.





	Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.	